SHORT · DATA LAKE

What is a data lake?

A data lake is cheap cloud storage that holds raw files in any format. You store first and decide later.

Key points

- CSV, JSON, images, and logs can all sit side by side.
- It is flexible and low cost, but a plain data lake has no built-in rules.
- Nothing stops a bad file, and it is hard to know which version is correct.
- That is why the lakehouse exists: Delta Lake and Unity Catalog add reliable tables and governance on top of the same cheap storage.

Tiny example

Write a few rows as raw JSON files into a volume, then list them. You get files, but no table and no history. In Databricks Free Edition, click New, then Notebook, and run each cell with Shift + Enter. No live output is recorded here.

```
spark.sql("CREATE VOLUME IF NOT EXISTS workspace.default.raw")
path = "/Volumes/workspace/default/raw/rows"

spark.range(3).write.mode("overwrite").json(path)
display(dbutils.fs.ls(path))   # raw JSON files: no table, no history
```

Misconception check

> A data lake is not the same as a lakehouse. The lake is the storage. The lakehouse adds reliable tables and governance on top of it.

Remember

A data lake stores everything cheaply, and the lakehouse makes it reliable.

The Short says you can try this in Databricks Free Edition, which includes the workspace catalog and Unity Catalog volumes. No paid features are needed.